

FINAL YEAR PROJECT INTERIM REPORT

P 001

HOSPITAL MANAGEMENT SYSTEM: A UNIFIED DIGITAL PLATFORM FOR PATIENT CARE, STAFF COORDINATION, AND ADMINISTRATIVE OPERATIONS

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**BACHELOR OF COMPUTER SCIENCE B.CS (HONS) SOFTWARE
ENGINEERING**

FEBRUARY 2026

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BY

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PROJECT INTERIM REPORT SUBMITTED IN PARTIAL FULFILMENT OF THE
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I hereby declare that the work has been done by myself and no portion of the work contained in this report has been submitted in support of any application for any other degree or qualification of this or any other university or institution of learning.

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ABSTRACT

Hospital management remains one of the most operationally complex domains in public service, requiring the coordination of clinical care, administrative workflows, medication dispensing, and financial processing across multiple departments and stakeholders. Existing systems in many healthcare institutions — particularly those serving mid-sized regional hospitals — continue to rely on fragmented, paper-based or siloed digital tools that result in data inconsistency, scheduling conflicts, delayed care delivery, and poor patient experience.

This project proposes a unified Hospital Management System (HMS) designed to digitise and integrate the core workflows of a hospital environment across six primary actors: Patient, Doctor, Nurse, Receptionist, Admin, and Pharmacist. The system enables end-to-end management of patient registration, appointment scheduling, electronic medical records (EMR), ward and bed management, prescription dispensing, billing, and operational reporting through a single, role-based web platform.

The system is built on a modular monolith architecture using Laravel for the RESTful backend and React for the web frontend. Patient data is stored in a relational MySQL database with enforced referential integrity. Real-time notifications are delivered via WebSockets to ensure time-critical updates — such as emergency admissions and lab results — reach the relevant staff immediately. A questionnaire-based requirements study conducted among 60 respondents (patients, clinical staff, and administrative personnel) confirmed significant pain points in appointment management, record accessibility, and inter-departmental communication, validating the need for an integrated HMS.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description
HMS	Hospital Management System
EMR	Electronic Medical Record
API	Application Programming Interface
ERD	Entity-Relationship Diagram
UML	Unified Modeling Language
UI	User Interface
OPD	Outpatient Department
IPD	Inpatient Department
RBAC	Role-Based Access Control

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CHAPTER 1: INTRODUCTION

1.1 Overview

Healthcare delivery depends on the accurate, timely, and coordinated exchange of information across multiple departments and professional roles. A hospital — even a mid-sized regional facility — must simultaneously manage outpatient appointments, inpatient admissions, clinical diagnoses, medication dispensing, laboratory results, billing, and staff scheduling. When these workflows operate in isolation — supported by paper records, standalone spreadsheets, or disconnected departmental software — the consequences include duplicated data entry, delayed treatment decisions, billing errors, and diminished patient trust.

The Hospital Management System (HMS) proposed in this project addresses these challenges by providing a unified, web-based digital platform that integrates the core operational and clinical workflows of a hospital. The system serves six distinct actor roles: Patient, Doctor, Nurse, Receptionist, Admin, and Pharmacist. Each role accesses a tailored interface that exposes only the functionality and data relevant to their responsibilities, governed by a role-based access control (RBAC) model.

From the patient's perspective, the system provides a self-service portal for appointment booking, medical history review, prescription access, and bill payment. Doctors interact with an electronic medical record (EMR) interface to log diagnoses, order prescriptions, and review patient history across visits. Nurses manage ward assignments, record patient vitals, and coordinate care handoffs. Receptionists handle patient registration and appointment scheduling at the front desk. Pharmacists receive and fulfill electronic prescriptions and manage medication inventory. Administrators oversee user management, departmental configuration, and operational reporting.

The platform is built using a modular monolith architecture — Laravel on the backend and React on the frontend — ensuring a clear separation of concerns while maintaining deployment simplicity appropriate for a student-phase project. Real-time features are implemented via WebSockets, and the system is backed by a relational MySQL database with a carefully designed schema that enforces data integrity across all inter-departmental relationships.

1.2 Problem Statement

Hospitals face a fundamental operational challenge: clinical care and administrative management generate high volumes of interconnected data that must flow accurately and promptly between departments, yet most mid-tier hospitals lack the integrated systems to support this. The consequences of fragmentation are observable across multiple dimensions.

Patient registration and appointment scheduling remain largely manual or semi-automated in many facilities. Receptionists record appointments in spreadsheets or paper logs, creating a high risk of double-bookings, missed appointments, and scheduling conflicts between doctors' availability and patient demand. Patients frequently arrive without confirmed slots, or their records cannot be retrieved quickly at the point of care.

Medical records present perhaps the most critical challenge. When a patient's history is stored across paper files, departmental silos, or incompatible legacy systems, attending doctors cannot make fully informed clinical decisions. Drug interactions may go undetected because prescription history is unavailable at the point of ordering. Lab results may not reach the treating physician until after the patient has been discharged.

The pharmacy represents another pain point. Handwritten or verbally communicated prescriptions introduce transcription errors and delays. Pharmacists cannot verify dosage histories, check for contraindications, or track inventory accurately when prescription data is not digitally integrated with the clinical record.

Billing and financial reconciliation suffer similarly. When patient data, services rendered, and insurance information exist in disconnected systems, billing errors multiply, payment delays increase, and the administrative burden on staff escalates. Patients experience confusion and distrust when bills do not clearly reflect the services they received.

Finally, administrators have limited visibility into operational performance. Without consolidated reporting across departments, identifying bottlenecks — such as consistently overbooked doctors, high bed occupancy in specific wards, or pharmacy stock shortfalls — requires laborious manual aggregation of data that may already be out of date.

These compounding inefficiencies motivate the design of an integrated Hospital Management System that treats the hospital as a connected system of roles, workflows, and shared data — rather than a collection of independent departments.

1.3 Project Objectives

The Hospital Management System aims to achieve the following objectives:

1. **Unified Patient Record Management:** Design and implement a centralised Electronic Medical Record (EMR) system that stores patient demographics, visit history, diagnoses, prescriptions, lab results, and billing data in a single relational database accessible to all authorised roles.
2. **Appointment and Scheduling Management:** Develop a scheduling module that allows patients to book appointments online and enables receptionists to

manage doctor availability, appointment slots, and walk-in registrations from a single interface.

3. **Role-Based Access Control (RBAC):** Implement a secure authentication and authorisation system that restricts data access and system functionality according to each user's role — Patient, Doctor, Nurse, Receptionist, Admin, or Pharmacist.
4. **Electronic Prescription and Pharmacy Integration:** Create a prescription workflow in which doctors issue electronic prescriptions that are immediately available to pharmacists for dispensing, with automatic inventory deduction and contraindication alerts.
5. **Ward and Bed Management:** Build a module enabling nurses and administrators to track bed occupancy, manage patient admissions and discharges, and assign ward staff to patients in real time.
6. **Automated Billing and Payment:** Implement a billing module that automatically generates invoices based on services rendered, supports multiple payment modes, and provides patients with itemised bills through their self-service portal.
7. **Real-Time Notifications:** Integrate a WebSocket-based notification system to deliver time-critical alerts — such as new appointment confirmations, emergency admissions, prescription readiness, and lab result availability — to the relevant role without page refresh.
8. **Operational Reporting and Analytics:** Provide administrators with a reporting dashboard presenting key operational metrics: daily patient throughput, departmental workload, bed occupancy rates, prescription volumes, and revenue summaries.
9. **Software Engineering Documentation:** Produce complete and compliant Software Engineering documentation including requirements specifications, UML diagrams, system architecture, database design, and testing plans in accordance with MMU FCI FYP standards.

1.4 Project Scope

The Hospital Management System is a web-based application targeting hospital environments of small-to-medium scale. The system operates across six defined user roles and covers the following functional domains within scope:

Patient Management: Registration, login, profile management, appointment booking, medical record viewing, prescription retrieval, and bill payment via the patient portal.

Outpatient Department (OPD): Doctor availability management, appointment scheduling by receptionists, patient queue management, and consultation recording by doctors.

Inpatient Department (IPD): Bed and ward assignment, admission and discharge management, daily nursing notes, and vitals recording.

Pharmacy: Electronic prescription receipt from doctors, medication dispensing, inventory tracking, and low-stock alerts.

Billing: Automatic invoice generation tied to consultation, prescription, lab, and admission records; payment processing; and receipt generation.

Administration: User account creation and role assignment, department management, and operational dashboard reporting.

Out of scope for this phase: - Laboratory Information System (LIS) integration with external lab equipment - Insurance claim submission and third-party payer integration - Telemedicine or video consultation features - Mobile application (native iOS/Android) - Multi-hospital or multi-branch deployment

The system is developed as a single-institution platform targeting one hospital environment. Scalability considerations are documented but multi-tenancy is deferred to a future phase.

1.5 Delivery

The Hospital Management System is expected to deliver measurable improvements across the operational and clinical dimensions of hospital management.

From a clinical perspective, the availability of a unified EMR ensures that every treating clinician — regardless of department — accesses the same patient history at the point of care. This reduces the risk of redundant tests, drug interaction errors, and information loss during patient handoffs between departments or care episodes. Doctors are expected to spend less time locating records and more time on clinical decision-making.

Operationally, the scheduling module is expected to eliminate appointment double-bookings and reduce patient wait times by enabling accurate, real-time management of doctor availability. Receptionists will be able to confirm, reschedule, or cancel appointments in seconds rather than minutes, and patients will receive automated confirmation and reminder notifications.

The pharmacy integration is expected to reduce prescription errors by removing handwritten orders from the dispensing workflow. Pharmacists will benefit from automatic inventory deduction and low-stock alerts, reducing the likelihood of stockouts for critical medications.

The billing module will reduce administrative reconciliation effort by automatically linking services to invoices. Patients will receive transparent, itemised bills through their portal, reducing billing disputes and improving trust in the institution.

For administrators, the reporting dashboard provides a live view of hospital performance without requiring manual data extraction. This enables proactive management of departmental bottlenecks, staffing adjustments, and resource allocation decisions.

In aggregate, the HMS is expected to improve the quality and efficiency of care delivery, reduce administrative overhead, and establish a digital foundation that can be extended with additional modules — such as laboratory integration, telemedicine, or insurance management — in subsequent project phases.

CHAPTER 2: BACKGROUND STUDY

2.1 Overview

This chapter reviews the academic literature and existing systems relevant to the development of a Hospital Management System. The review is structured around the key thematic challenges identified in the problem statement: patient record management, appointment scheduling, pharmacy integration, billing, and inter-departmental coordination. Following the literature review, five commercially deployed or open-source HMS platforms are evaluated to identify their strengths, limitations, and the gaps that the proposed system aims to address.

2.2 Literature Review

2.2.1 Electronic Medical Records and Clinical Outcome Improvement

The transition from paper-based to electronic medical records has been extensively studied. Campanella et al. (2016) conducted a systematic review of 14 studies and found that EMR adoption was associated with statistically significant improvements in adherence to clinical guidelines and reductions in medication errors. However, the review noted that benefits were contingent on system usability and staff training quality — poorly designed EMR interfaces introduced new cognitive burdens that partially offset efficiency gains.

Boonstra and Broekhuis (2010) identified resistance to EMR adoption among clinical staff as a primary implementation challenge, categorising barriers into financial, technical, time-related, psychological, and social dimensions. Their framework remains relevant to this project: the proposed HMS must prioritise interface simplicity and role-specific workflows to minimise the learning curve and reduce resistance to adoption.

2.2.2 Appointment Scheduling and Patient Wait Time Reduction

Scheduling inefficiencies represent one of the most quantifiable inefficiencies in outpatient hospital management. Cayirli and Veral (2003) provided a comprehensive review of outpatient scheduling models, demonstrating that computerised scheduling systems consistently reduced patient wait times and improved resource utilisation compared to manual appointment booking. Their analysis identified doctor no-show prediction and patient arrival variance as key variables that scheduling systems must accommodate.

More recently, Ahmadi-Javid et al. (2017) reviewed over 200 papers on outpatient appointment scheduling and concluded that systems which integrate

real-time queue visibility with dynamic slot allocation yield the greatest reduction in patient wait times. The proposed HMS incorporates real-time doctor availability management and appointment slot locking to address these findings.

2.2.3 Pharmacy Information Systems and Medication Safety

Bates et al. (1999) established foundational evidence that computerised physician order entry (CPOE) with pharmacy integration reduces serious medication errors by 55%. Their study at Brigham and Women's Hospital demonstrated that electronic prescription systems — by enforcing structured data entry and enabling automatic contraindication checking — eliminate the transcription errors inherent to handwritten orders.

Subsequent work by Kaushal et al. (2003) extended these findings to cost-benefit analysis, demonstrating that pharmacy information system implementations yielded net cost savings within three years through reduced adverse drug events and associated treatment costs. These findings directly motivate the electronic prescription and pharmacy integration module in the proposed HMS.

2.2.4 Hospital Billing Systems and Revenue Cycle Management

Healthcare billing represents a major source of administrative waste. Himmelstein et al. (2014) estimated that billing and insurance-related activities consume 34.2% of hospital revenue in the United States, with significant inefficiency attributable to manual data entry, claim rework, and denial management. While the regulatory context differs across countries, the core inefficiency — disconnected billing data — is universal.

Adler-Milstein et al. (2013) found that hospitals with integrated billing and clinical systems experienced significantly lower claim denial rates and shorter revenue cycles compared to those relying on manual reconciliation. The proposed HMS addresses this by auto-generating invoices directly from recorded clinical events, eliminating manual transcription between the clinical and billing domains.

2.2.5 Role-Based Access Control in Healthcare Systems

Patient data privacy and regulatory compliance require that healthcare information systems implement strict access controls. Ferraiolo et al. (2001) formalised the RBAC model as the standard framework for enterprise access control, demonstrating its appropriateness for environments with clearly defined professional roles — such as hospitals — where permissions are determined by job function rather than individual identity.

In the healthcare context, Liu et al. (2012) validated RBAC as the dominant access control model in hospital information systems, noting that properly implemented RBAC reduces unauthorised data access incidents and simplifies compliance auditing. The proposed HMS implements RBAC across all six actor roles, ensuring that each user accesses only the data and functions appropriate to their professional responsibilities.

2.2.6 Summary of Research Gaps

The literature establishes strong evidence for the benefits of integrated HMS solutions but also reveals several recurring gaps in existing deployments:

- Most studied systems are deployed in large, well-resourced hospitals. Evidence for mid-tier hospital implementations is limited.
- Pharmacy integration with real-time contraindication checking is technically complex and often simplified or omitted in resource-constrained systems.
- Patient-facing portals are frequently designed as afterthoughts rather than primary interfaces, resulting in low adoption.
- Real-time inter-departmental notification remains under-studied as a factor in care coordination quality.
- Open-source HMS platforms exist but require significant technical customisation that many hospitals cannot support internally.

These gaps collectively motivate the proposed HMS, which targets mid-tier hospital environments with a practical, deployable architecture and a patient-facing portal designed as a first-class component.

2.3 Existing Systems

2.3.1 Epic Systems

Epic is the dominant electronic health record and hospital management platform in the United States, used by many of the largest health systems globally. It provides comprehensive clinical, administrative, and financial management capabilities through an integrated suite of modules covering EMR, scheduling, billing, pharmacy, laboratory, and patient portal (MyChart).

Epic's strength lies in its completeness and depth of clinical functionality. It supports complex clinical workflows, interoperability standards (HL7, FHIR), and sophisticated reporting. However, Epic is designed for large health systems and enterprise deployments. Licensing costs are prohibitive for mid-tier hospitals, implementation timelines typically span 18–36 months, and the system requires dedicated IT teams for ongoing maintenance.

Table 1: Epic Systems Feature Summary

Feature	Epic
EMR	Yes — comprehensive
Appointment Scheduling	Yes
Pharmacy Integration	Yes
Billing	Yes
Patient Portal	Yes (MyChart)
Real-Time Notifications	Yes
Open Source	No
Target Scale	Large enterprise

Limitations for this project's context: - Prohibitively expensive for mid-tier hospitals - Requires large dedicated IT teams - Not suitable for academic or prototype deployment

2.3.2 Cerner (Oracle Health)

Cerner, now part of Oracle Health following the 2022 acquisition, is Epic's primary competitor in the enterprise HMS market. It provides similar breadth of clinical and administrative functionality, with particular strength in interoperability and cloud deployment options. Like Epic, Cerner targets large hospital systems and academic medical centres.

Cerner's acquisition by Oracle has introduced cloud-native architecture options, which represent a technical advancement. However, the platform remains enterprise-focused, with similarly high costs and implementation complexity.

Table 2: Cerner Feature Summary

Feature	Cerner
EMR	Yes
Appointment Scheduling	Yes
Pharmacy Integration	Yes
Billing	Yes
Patient Portal	Yes
Real-Time Notifications	Yes
Open Source	No
Target Scale	Large enterprise

2.3.3 OpenMRS

OpenMRS is an open-source medical record system designed for deployment in resource-limited settings, particularly in developing countries. It was developed collaboratively by a global community and is used in healthcare facilities across Africa, Asia, and Latin America. OpenMRS provides a configurable data model that can be adapted to local clinical workflows.

OpenMRS is notable for its accessibility and community support. It is freely available and can be deployed with modest infrastructure. However, it is primarily an EMR system — not a full HMS. It lacks native appointment scheduling, pharmacy management, billing, and administrative reporting modules. These must be added through third-party modules or custom development, creating integration and maintenance challenges.

Table 3: OpenMRS Feature Summary

Feature	OpenMRS
EMR	Yes
Appointment Scheduling	Limited (module)
Pharmacy Integration	Limited (module)
Billing	No (third-party)
Patient Portal	No
Real-Time Notifications	No
Open Source	Yes
Target Scale	Small to mid-tier

2.3.4 HospitalRun

HospitalRun is another open-source HMS designed for low-resource environments. Unlike OpenMRS, it aims to be a more complete HMS covering patient management, appointments, pharmacy, and billing. It features an offline-first architecture that allows operation in environments with unreliable internet connectivity.

While HospitalRun's scope is closer to the proposed system, its development has been inconsistent — the project experienced a significant hiatus between 2019 and 2022 — and the current codebase lacks maturity in several modules. The patient portal, in particular, is underdeveloped, and the reporting capabilities are basic.

Table 4: HospitalRun Feature Summary

Feature	HospitalRun
EMR	Yes
Appointment Scheduling	Yes
Pharmacy Integration	Yes
Billing	Yes
Patient Portal	No
Real-Time Notifications	No
Open Source	Yes
Target Scale	Small to mid-tier

2.3.5 Practo

Practo is a commercial healthcare platform widely used in South and Southeast Asia. It combines a patient-facing appointment booking marketplace with clinic and hospital management tools. Practo's patient portal is its strongest feature — it allows patients to search for doctors, book appointments, access prescriptions, and view health records through a polished mobile and web interface.

However, Practo's HMS backend is optimised for small clinics and private practices rather than full hospital environments. It lacks ward and bed management, inpatient workflows, and the depth of pharmacy and billing integration required for a multi-department hospital. Its reporting is limited to practice-level analytics.

Table 5: Practo Feature Summary

Feature	Practo
EMR	Partial
Appointment Scheduling	Yes (marketplace)
Pharmacy Integration	Partial
Billing	Partial
Patient Portal	Yes — strong
Real-Time Notifications	Yes
Open Source	No
Target Scale	Clinics / small hospitals

2.3.6 Comparison of Existing Systems

Table 6: Comparison of Existing HMS Platforms

Feature	Epic	Cerner	Open-MRS	Hospital-Run	Practo	Proposed HMS
EMR	Yes	Yes	Yes	Yes	Partial	Yes
Appointment Scheduling	Yes	Yes	Limited	Yes	Yes	Yes
Pharmacy Integration	Yes	Yes	Limited	Yes	Partial	Yes
Billing	Yes	Yes	No	Yes	Partial	Yes
Patient Portal	Yes	Yes	No	No	Yes	Yes
Real-Time Notifications	Yes	Yes	No	No	Yes	Yes
Ward / Bed Management	Yes	Yes	No	No	No	Yes
RBAC (6 roles)	Yes	Yes	Partial	Partial	No	Yes
Open Source	No	No	Yes	Yes	No	Yes (academic)
Mid-tier Hospital Fit	Poor	Poor	Moderate	Moderate	Poor	Strong

The comparison reveals that enterprise platforms (Epic, Cerner) provide full functionality but are inaccessible to mid-tier hospitals due to cost and complexity. Open-source alternatives (OpenMRS, HospitalRun) are accessible but functionally incomplete. Practo excels in patient-facing features but lacks inpatient and administrative depth. The proposed HMS targets the gap between these categories: a complete, mid-tier HMS with a strong patient portal, real-time notifications, and full role-based functionality, deployable without enterprise-level infrastructure.

CHAPTER 3: REQUIREMENTS ANALYSIS

3.1 Overview

This chapter presents the requirements analysis for the Hospital Management System. Requirements were gathered through a structured questionnaire distributed to 60 respondents across three stakeholder groups: hospital patients (25), clinical staff including doctors and nurses (20), and administrative staff including receptionists and pharmacists (15). The questionnaire targeted real-world pain points in hospital workflows to validate the system's design priorities and derive functional and non-functional requirements grounded in user need.

Following the questionnaire analysis, functional requirements are presented by module and actor, and non-functional requirements are specified. The chapter concludes with a Use Case Diagram capturing the interactions between the six system actors and the HMS.

3.2 Questionnaire

3.2.1 Question 1: Difficulty Scheduling Hospital Appointments

This question assessed whether patients and staff experience difficulty with the current appointment scheduling process.

Q1: I find it difficult to schedule a hospital appoin

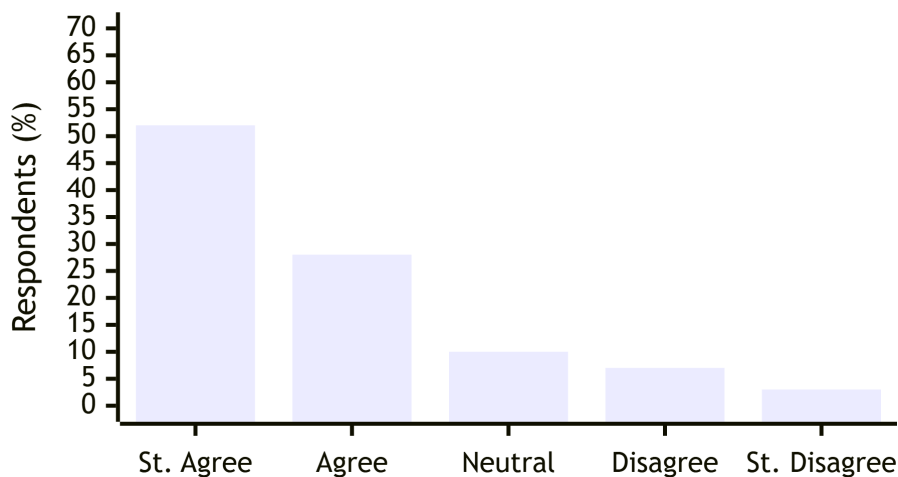


Figure 1: Q1: I find it difficult to schedule a hospital appointment

Finding: 80% of respondents agreed or strongly agreed that scheduling appointments is difficult. This strongly validates the need for an integrated scheduling module accessible to both patients and receptionists.

3.2.2 Question 2: Paper-Based Records Cause Delays

This question examined whether paper-based or disconnected records cause delays in patient care.

Q2: Paper-based records cause delays in receiving

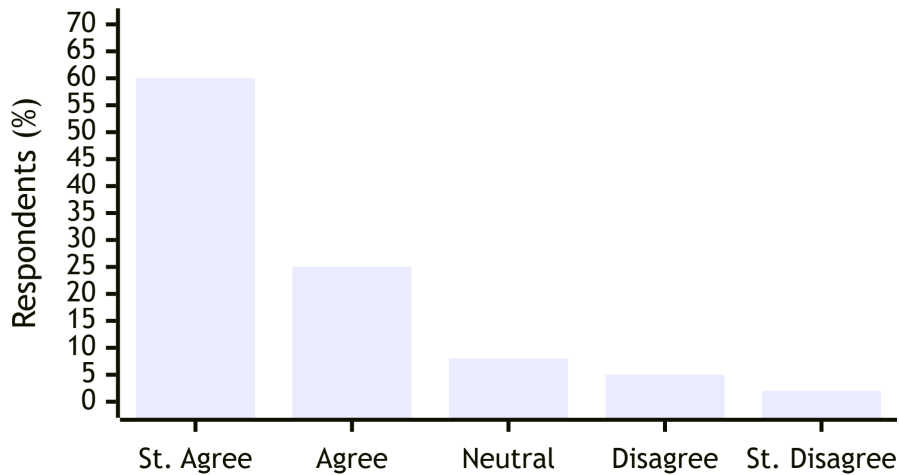


Figure 2: Q2: Paper-based records cause delays in receiving care

Finding: 85% agreed that fragmented records delay care. This is the strongest signal in the questionnaire and directly motivates the centralised EMR as the foundation of the HMS.

3.2.3 Question 3: Difficulty Accessing Personal Medical History

This question explored whether patients can easily access their own medical records.

Q3: I can easily access my own medical records w/

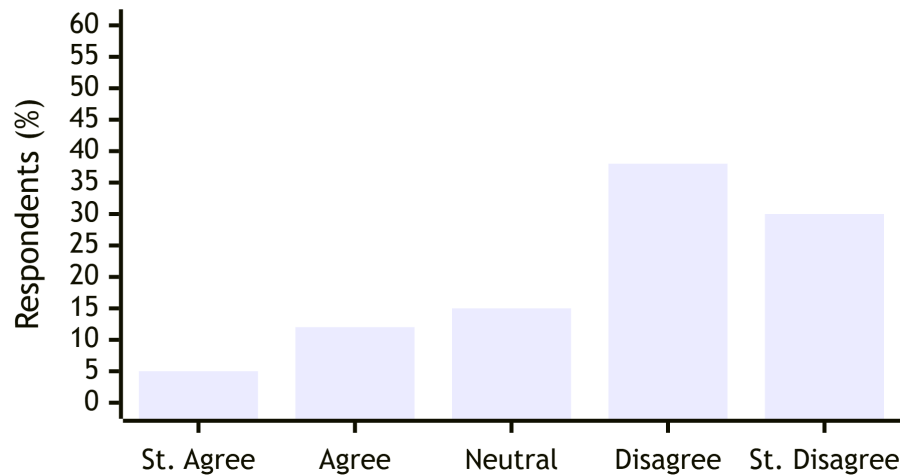


Figure 3: Q3: I can easily access my own medical records when needed

Finding: 68% of patients disagreed or strongly disagreed that their records are easily accessible. This validates the patient portal as a priority feature, not an optional add-on.

3.2.4 Question 4: Prescription Errors Due to Handwritten Orders

This question targeted clinical and pharmacy staff regarding medication errors from handwritten prescriptions.

Q4: Handwritten prescriptions have led to dispens

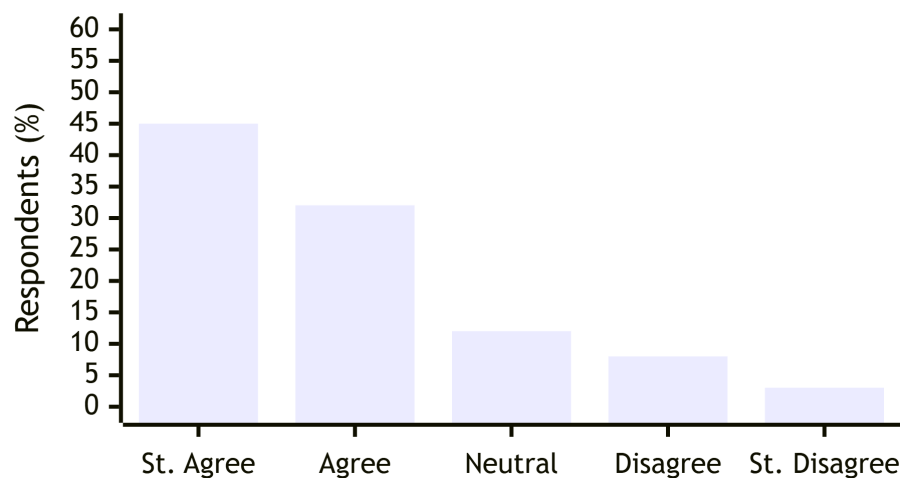


Figure 4: Q4: Handwritten prescriptions have led to dispensing errors

Finding: 77% of clinical and pharmacy respondents confirmed that handwritten prescriptions cause dispensing errors. This directly supports the electronic prescription module and pharmacy integration.

3.2.5 Question 5: Billing Errors and Transparency

This question assessed patient experience with hospital billing accuracy.

Q5: I have experienced billing errors or unclear invoices

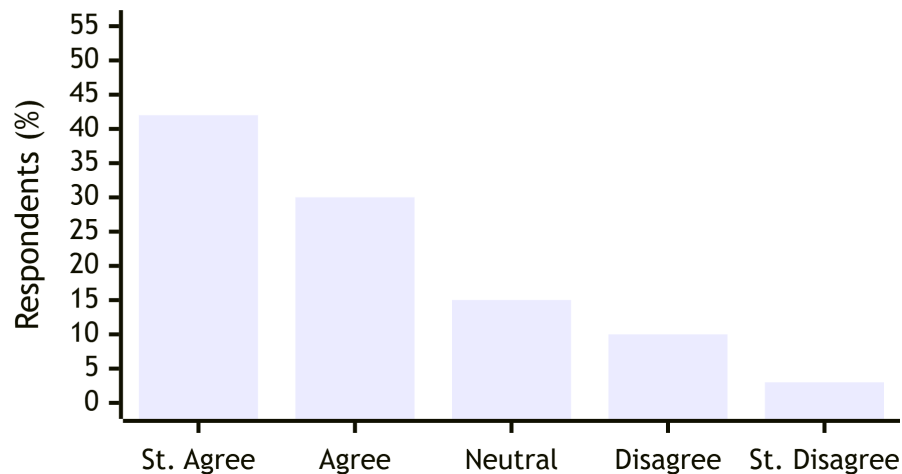


Figure 5: Q5: I have experienced billing errors or unclear invoices

Finding: 72% of patients reported experiencing billing errors or receiving invoices they could not understand. This validates the automated billing module with itemised, patient-visible invoices.

3.2.6 Question 6: Usefulness of a Patient Self-Service Portal

This question assessed patient interest in a self-service digital portal.

Q6: A patient portal for appointments and records

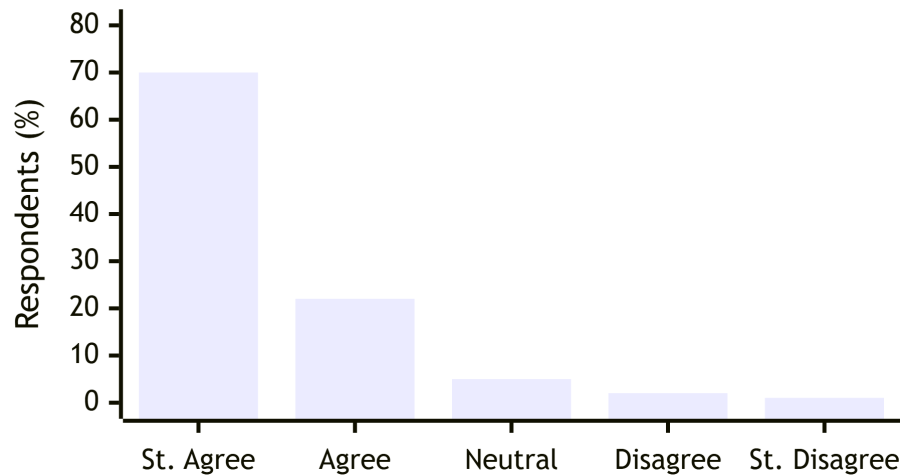


Figure 6: Q6: A patient portal for appointments and records would be useful

Finding: 92% expressed strong interest in a patient portal. This is the highest agreement rate in the questionnaire, confirming that patient self-service is a primary deliverable.

3.2.7 Question 7: Ward and Bed Management Challenges

This question targeted nursing and administrative staff on challenges with ward management.

Q7: Managing bed assignments and ward status is a daily challenge

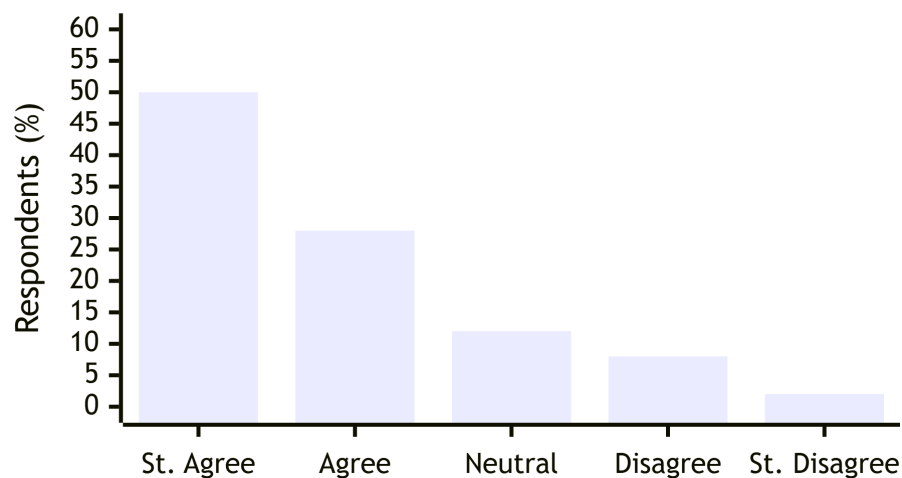


Figure 7: Q7: Managing bed assignments and ward status is a daily challenge

Finding: 78% of nursing and admin staff confirmed ward management as a daily operational challenge, validating the bed and ward management module.

3.2.8 Question 8: Real-Time Notifications for Critical Updates

This question explored whether staff receive critical information promptly.

Q8: I often receive critical patient updates too late

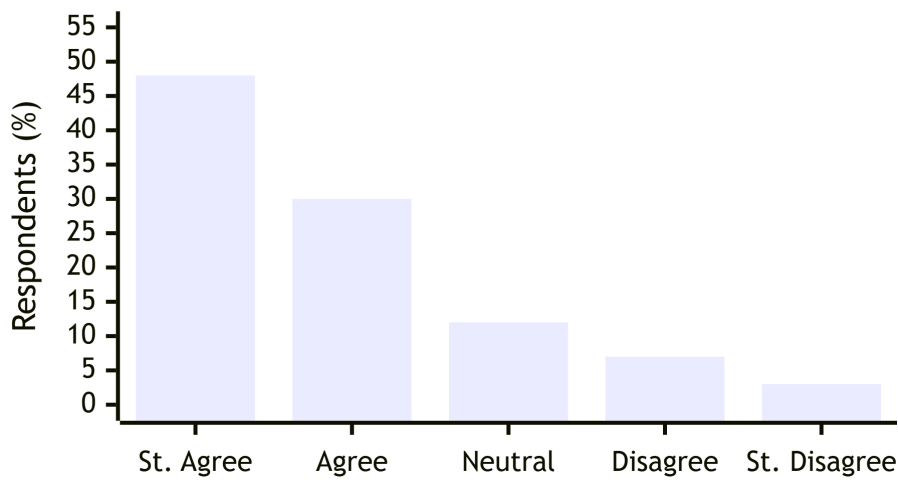


Figure 8: Q8: I often receive critical patient updates too late

Finding: 78% of staff confirmed that critical updates — such as new admissions, lab results, or emergency changes — arrive too slowly. This validates the Web-Socket-based real-time notification system.

3.2.9 Question 9: Administrative Reporting and Data Visibility

This question targeted administrative staff on access to operational data.

Q9: I have adequate visibility into hospital operati

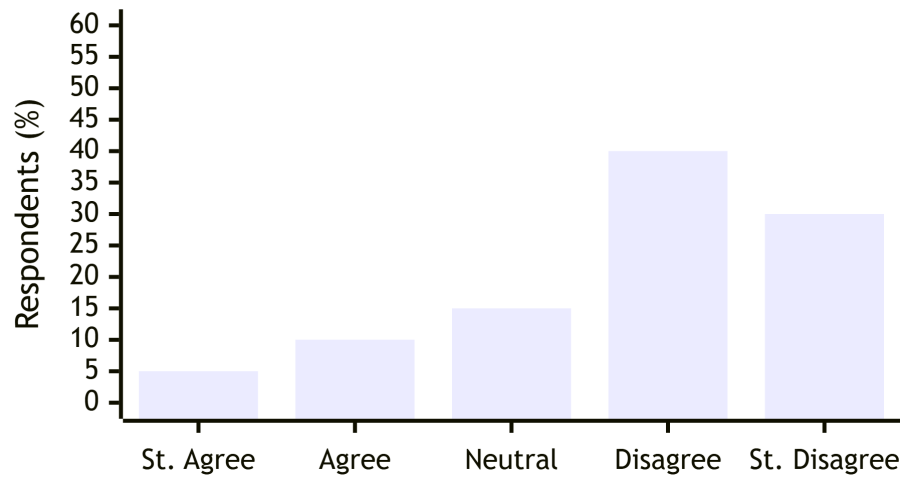


Figure 9: Q9: I have adequate visibility into hospital operational data

Finding: 70% of administrators and senior staff reported inadequate operational data visibility. This validates the reporting dashboard as a meaningful administrative feature.

3.2.10 Question 10: Overall Need for an Integrated HMS

This question assessed the overall perceived need for an integrated HMS platform.

Q10: An integrated HMS would significantly improv

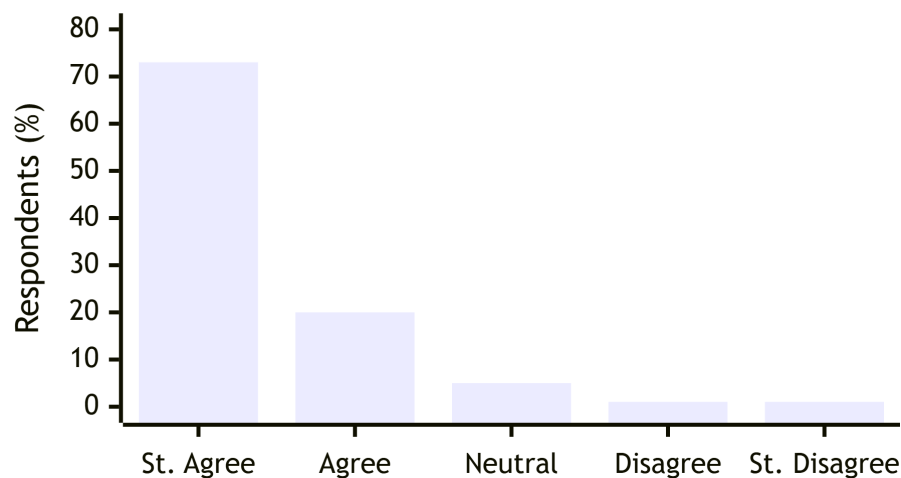


Figure 10: Q10: An integrated HMS would significantly improve hospital operations

Finding: 93% of all respondents agreed that an integrated HMS would significantly improve hospital operations — the strongest cross-stakeholder consensus in the entire questionnaire. This provides broad validation for the project as a whole.

3.3 Functional Requirements

Functional requirements are organised by system module. Each module maps to one or more of the six system actors.

3.3.1 F1 — User Authentication and Role Management

Table 7: F1 – Authentication and Role Management

Req ID	Requirement	Description
F1.1	User Registration	System shall allow new users to register with name, email, password, and assigned role.
F1.2	User Login	System shall authenticate users via email and password and issue session tokens.
F1.3	Role-Based Access Control	System shall restrict access to modules and data based on the authenticated user's role.
F1.4	Password Reset	System shall allow users to reset their password via email verification.
F1.5	Session Management	System shall maintain secure sessions and auto-expire inactive sessions after 30 minutes.
F1.6	Admin User Management	Admin shall create, deactivate, and reassign user accounts and roles.

3.3.2 F2 — Patient Management

Table 8: F2 – Patient Management

Req ID	Requirement	Description
F2.1	Patient Registration	Receptionist shall register new patients with demographic, contact, and insurance information.
F2.2	Patient Search	All clinical roles shall search for patients by name, ID, or phone number.
F2.3	Patient Profile	System shall maintain a persistent patient profile linking all visits, diagnoses, prescriptions, and bills.
F2.4	Patient Portal Access	Patients shall log in to view their appointments, medical records, prescriptions, and invoices.
F2.5	Medical History View	Doctors and nurses shall view a patient's full visit history, diagnoses, and prescriptions.

3.3.3 F3 — Appointment Scheduling

Table 9: F3 – Appointment Scheduling

Req ID	Requirement	Description
F3.1	Book Appointment (Patient)	Patients shall book appointments by selecting department, doctor, and available time slot.
F3.2	Book Appointment (Receptionist)	Receptionists shall schedule appointments on behalf of walk-in patients.
F3.3	Doctor Availability Management	Doctors shall define and update their weekly availability.
F3.4	Appointment Confirmation	System shall send confirmation notifications to patients upon booking.
F3.5	Appointment Cancellation	Patients and receptionists shall cancel appointments with automated notification.
F3.6	Appointment Reminder	System shall send reminder notifications to patients 24 hours before their appointment.
F3.7	Daily Queue View	Receptionists and doctors shall view the daily appointment queue per doctor.

3.3.4 F4 — Electronic Medical Records (EMR) and Clinical Workflow

Table 10: F4 – EMR and Clinical Workflow

Req ID	Requirement	Description
F4.1	Record Consultation	Doctors shall record consultation notes, diagnosis, and treatment plan per patient visit.
F4.2	Issue Prescription	Doctors shall issue electronic prescriptions linked to the patient record and forwarded to pharmacy.
F4.3	Record Vitals	Nurses shall record patient vitals (blood pressure, temperature, weight, oxygen level) per visit.
F4.4	Nursing Notes	Nurses shall add shift-based care notes to the patient's active admission record.
F4.5	Lab Order	Doctors shall issue lab test orders linked to the patient record.
F4.6	Lab Result Upload	System shall allow lab results to be uploaded and attached to the patient record with doctor notification.

3.3.5 F5 — Ward and Bed Management

Table 11: F5 – Ward and Bed Management

Req ID	Requirement	Description
F5.1	Bed Availability View	Nurses and receptionists shall view real-time bed availability by ward.
F5.2	Admit Patient	Receptionists shall admit a patient to a ward and assign a bed.
F5.3	Discharge Patient	Doctors shall authorise patient discharge, triggering billing finalisation.
F5.4	Transfer Patient	Nurses shall transfer a patient between wards with bed status updated automatically.
F5.5	Ward Occupancy Report	Admin shall view ward occupancy rates by department and date range.

3.3.6 F6 — Pharmacy Management

Table 12: F6 – Pharmacy Management

Req ID	Requirement	Description
F6.1	Receive Prescription	Pharmacists shall receive electronic prescriptions issued by doctors in real time.
F6.2	Dispense Medication	Pharmacists shall mark prescriptions as dispensed, automatically deducting inventory.
F6.3	Inventory Management	Pharmacists shall manage medication stock levels and view current inventory.
F6.4	Low Stock Alert	System shall notify pharmacists when a medication's inventory falls below a defined threshold.
F6.5	Prescription History	Pharmacists shall view a patient's prescription history to check for repeat dispensing.

3.3.7 F7 — Billing and Payment

Table 13: F7 – Billing and Payment

Req ID	Requirement	Description
F7.1	Auto Invoice Generation	System shall automatically generate an invoice upon consultation, admission, prescription, or lab event.
F7.2	Itemised Bill View	Patients shall view an itemised bill listing each charged service in their portal.
F7.3	Payment Processing	Receptionists shall process payments and mark invoices as paid.
F7.4	Receipt Generation	System shall generate a payment receipt upon successful payment.
F7.5	Outstanding Balance Alert	System shall notify receptionists and patients of outstanding invoice balances.

3.3.8 F8 — Administration and Reporting

Table 14: F8 – Administration and Reporting

Req ID	Requirement	Description
F8.1	Department Management	Admin shall create and manage hospital departments and assign doctors to departments.
F8.2	Daily Patient Report	Admin shall generate a daily report of patient registrations, admissions, and discharges.
F8.3	Revenue Report	Admin shall generate revenue reports by date range, department, or doctor.
F8.4	Pharmacy Consumption Report	Admin shall view medication dispensing volumes and inventory valuation by period.
F8.5	System Audit Log	System shall log all significant user actions for security and compliance auditing.

3.4 Non-Functional Requirements

Table 15: Non-Functional Requirements

Req ID	Category	Requirement
NF1	Performance	System shall respond to any user action within 2 seconds under normal load.
NF2	Security	Passwords shall be hashed using bcrypt. All API endpoints shall require bearer token authentication.
NF3	Data Privacy	Patient data shall be accessible only to authorised roles per RBAC policy.
NF4	Availability	System shall maintain 99% uptime during hospital operating hours (6:00 AM – 12:00 AM).
NF5	Usability	Core workflows shall be completable in under 5 user interactions.
NF6	Compatibility	System shall function correctly on Chrome, Firefox, and Edge on desktop.
NF7	Scalability	System shall support up to 500 concurrent users without schema changes.
NF8	Data Integrity	Foreign key relationships shall be enforced at the database level. Cascading deletes applied only where clinically safe.
NF9	Audit Trail	Patient record operations (create, update, delete) shall generate an immutable audit log entry.
NF10	Maintainability	Backend shall use a modular monolith with separated service layers.

3.5 Use Case Diagram

The diagram below presents the HMS use case model. Six actors interact with the system across eight functional modules.

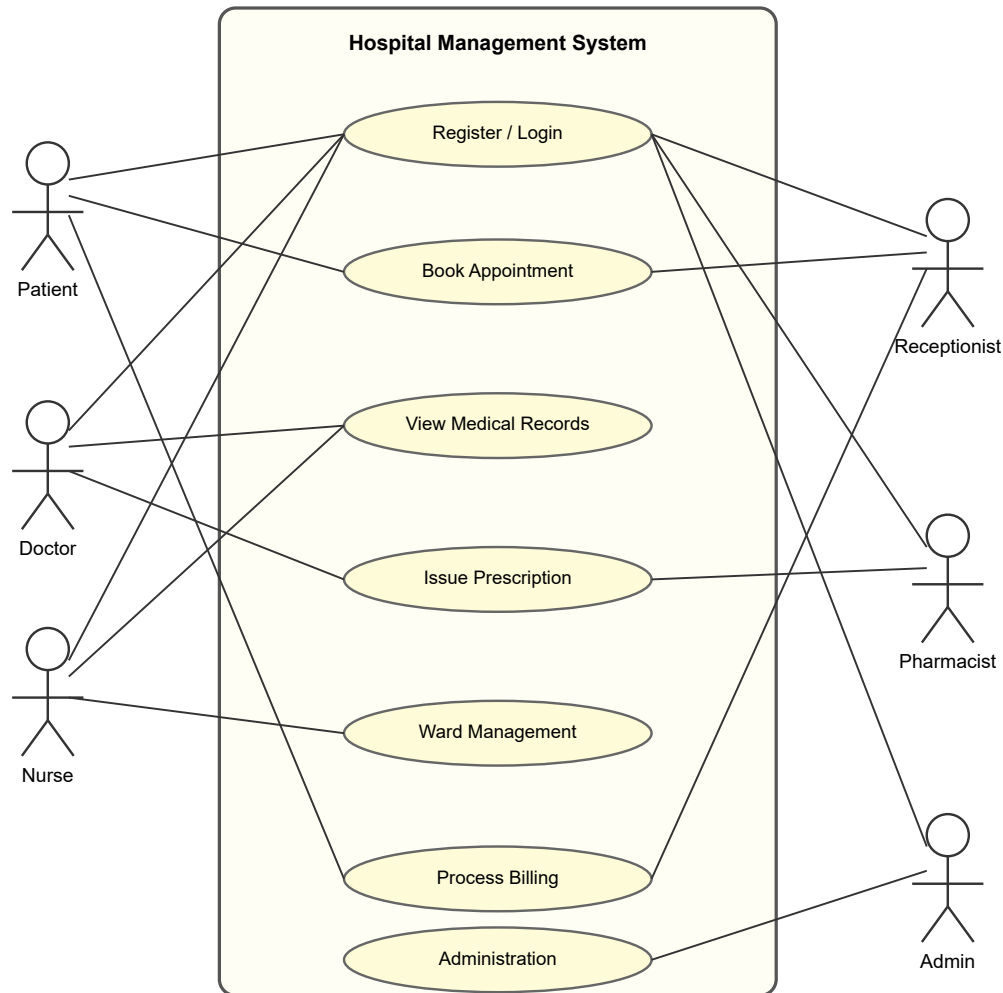


Figure 11: HMS Use Case Diagram

3.5.1 Actor Responsibilities Summary

Table 16: Actor Responsibility Summary

Actor	Primary Responsibilities
Patient	Self-registration, appointment booking, medical record viewing, invoice viewing
Doctor	Consultation recording, prescription issuance, lab orders, discharge authorisation
Nurse	Vitals recording, nursing notes, bed and ward management
Receptionist	Patient registration, appointment management, queue management, payment processing
Pharmacist	Prescription receipt and dispensing, inventory management
Admin	User management, department configuration, operational reporting

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APPENDIX A: GANTT CHART

Attach your Gantt chart here.

APPENDIX B: FYP I MEETING LOGS

Replace this text with your meeting logs. A minimum of six (6) meeting logs are required.

APPENDIX C: TURNITIN SIMILARITY INDEX PAGE

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